$f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$F : \mathbb{R}^n \longrightarrow \mathbb{R}^n$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots\ldots \frac{\partial f}{\partial x_n}(x) \right)$$ gradient

$$\nabla \cdot F(x) = \frac{\partial F_1}{\partial x_1}(x) + \ldots + \frac{\partial F_n}{\partial x_n}(x)$$ divergence

$$\nabla \times F(x) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{F_2}{\partial x} - \frac{F_1}{\partial y} \right)$$

$n = 3$

$$\Delta f = \frac{\partial^2 f}{\partial x_1^2} + \ldots + \frac{\partial^2 f}{\partial x_n^2}$$

$n = 3$

$\nabla \times \nabla f = 0$

$$\nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2}{\partial z \partial y}, \dots, \dots \right) = \vec{0}$$

$$\nabla \cdot (\nabla \times F) = 0$$

$$\nabla \cdot (-y_z + z_y, -z_x + x_z, -x_y + y_x) =$$

$$\cancel{z_{yx}} - \cancel{y_{zx}} + \cancel{x_{zy}} - \cancel{z_{xy}} + \cancel{y_{xz}} - \cancel{x_{yz}} = 0$$

cotomological complex, ke imaš duc

zapoledn: pušči je 0

$$\mathbb{R} \xrightarrow{\nabla} \mathbb{R}^3 \xrightarrow{\nabla \times} \mathbb{R}^3 \xrightarrow{\nabla \cdot} \mathbb{R}$$

$K = [0,1] \times [0,1] \subseteq \mathbb{R}^2$

$$\int_K \nabla \cdot \vec{F}\, dx\, dy = \int_{\partial K} \vec{F}\, d\vec{n}$$

normale

$$\int_0^1 \int_0^1 \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx\, dy = \int_0^1 dy \int_0^1 \frac{\partial F_1(x,y)}{\partial x} dx + \int_0^1 dx \int_0^1 \frac{\partial F_2(x,y)}{\partial y} dy$$

$$= \int_0^1 F_1(x,y)\Big|_0^1 dy + \int_0^1 F_2(x,y)\Big|_0^1 dx =$$

$$= \int_0^1 \left(F_1(1,y) - F(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$$\int_{\partial K} \vec{F}\, d\vec{n} = \int_0^1 \left(F_1(x,0), F_2(x,0) \right) \begin{pmatrix} dx \\ 0, -1 \end{pmatrix} dy + \dots$$

$$= \int_0^1 -F_2(x,0)\, dx + \dots = \int_0^1 \left(F_1(1,y) - F_1(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$u_{xy} = x^2 y \qquad u(x,0) = x \qquad u(0,y) = y^2$

$$u_x = \frac{1}{2}x^2y^2 + C(x)$$

$$u = \frac{1}{6}x^3y^2 + C(x) + D(y)$$

$$u(x,0) = C(x) + D(0) = x$$

$$u(0,y) = C(0) + D(y) = y^2$$

$$C(x) + D(y) = x + y^2$$

$C(0)$ in $D(0)$ inhlo dano za 0

$$u(x,y) = \frac{1}{6}x^3y^2 + x + y^2$$

$$u_{xy} + u_x = y e^{-y}$$

$$u(0,y) = 0$$

$$u(x,x) = 0$$

$$u_y(0,y) = 0$$

$$\frac{u(x,x)}{\partial y} = 0$$

integriramo po x

$$u_y + u = xye^{-y} + C(y)$$

v $(0,y) \rightsquigarrow$

$$0 + 0 = 0 + C(y) = 0$$

$$u_y + u = xye^{-y}$$

linearna diferencialna enačba

homogen del

$$u + u_y = 0$$

$$u = -\frac{\partial u}{\partial y} \quad dy = -\frac{du}{u} \rightsquigarrow y = -\ln u + C(x)$$

$$u = Ce^{-y}$$

partikularna

$$C'(y)e^{-y} = xye^{-y}$$

$$C'(y) = xy$$

$$C(x,y) = \frac{1}{2}xy^2$$

$$u = \frac{1}{2}xy^2e^{-y} + C(x)e^{-y}$$

$$u(0,y) = C(0)e^{-y} = 0$$

$$u(x,x) = \frac{1}{2}x^3e^{-x} + C(x)e^{-x} = 0$$

$$\Rightarrow C(x) = -\frac{1}{2}x^3$$

$$u(x,y) = \frac{1}{2}x\,e^{-x}(y^2 - x^2)$$

c)

$u_x + u_y = 0 \qquad u(x,0) = x^2$

$$u(x,y) = \sum_{i,j} a_{ij} x^i y^j$$

$$u(x,0) = \sum_i a_{i0} x^i = x^2 \Rightarrow a_{20} = 1$$

$a_{i0} = 0$ za $i \neq 2$

$$u_x = \sum_{i,j} a_{ij}\, i\, x^{i-1} y^j = a_{10} + \sum_{i,j\geq 0} a_{i+1\,j}\,(i+1) x^i y^j$$

$$u_y = \sum_{i,j} a_{ij}\, j\, x^j y^{j-1} = a_{01} + \sum_{i,j\geq 0} a_{i\,j+1}\,(j+1)\, y^j x^i$$

$$u_x + u_y = a_{01} + a_{10} + \sum_{i,j\geq 1} (a_{i+1,j}(i+1) + a_{i,j+1}(j+1)) x^i y^j = 0$$

$$\Rightarrow (a_{i+1})(i+1) + a_{i,j+1}(j+1) = 0 \text{ za } \forall\, xy$$

$a_{01} + a_{10} = 0 \Rightarrow a_{01} = -a_{10}$

$\Rightarrow a_{0j} = 0$ razen $a_{02} = -2$

$a_{2,1} = a_{1,2}$

$a_{11} = -a_{0,2}$

$a_{11} = -a_{2,0}$

$a_{1,0} + a_{01} = 0,\ a_{1,1} + a_{02} = 0 \Rightarrow a_{11} = 2$

j \ i	0	1	2	3		
0	a_{00}	0	1	0	0	...
1	0	-2	0	0		
2	1	0	0			
3	0	0				
	0					

$a_{3,0} = -a_{2,1}$

$a_{11} = -a_{02}$

$$u(x,y) = a_{00} + a_{2,0} x^2 + a_{11} xy + a_{02} y^2$$

$$= a_{00} + x^2 - xy + y^2$$

$$u(x,0) = a_{00} + x^2 \Rightarrow a_{00} = 0$$

$\Rightarrow u(x,y) = x^2 - xy + y^2 \qquad a_{2,0} = a_{11}$

Hmmm nekaj je narobe

$u(x,y) = x^2 - 2xy + y^2$

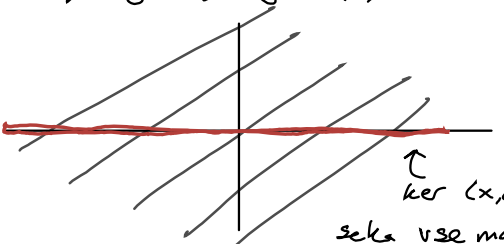
Pozabila sem množiti z $i+1$ in $j+1$ pri odvajanju

3. c)

$u_x + u_y = 0 \qquad u(x,0) = x^2 \quad \rightsquigarrow u(x,y) = (x-y)^2$

Naj bo $u(x,y) = g(x-y)$

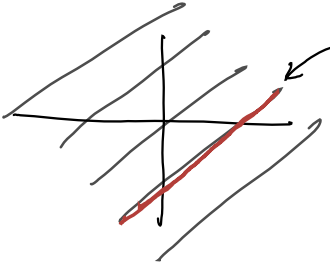
$\Rightarrow u_x + u_y = g'(x-y) - g'(x-y) = 0$



ker $(x,0)$ seka vse možnosti, je to validn način

Slab primer

$u(x,x) = x^2$



če imamo tak robni pogoj ne moremo dobiti rešitve

Ali neobstaja, ali pa jako je veliko

$$u_{xy} + y u_y + u = 1 \qquad u_x(x,0) = 1$$
$$u(x,x) = e^{-x^2}$$

$$y u_y + u = (yu)_y$$

$$u_{xy} + (yu)_y = 1 \rightsquigarrow u_x + yu = y + C(x)$$

$$V(x,0): \quad \overset{0}{\overset{\shortparallel}{u_x(x,0)}} + 0 \cdot 1 = 0 + C(x) \Rightarrow C(x) = 0$$

$$u_x + yu = y$$

homogeno:

$$u_x = -yu$$
$$\frac{\partial u}{u} = -y\,dx$$
$$u = D(y) e^{-yx}$$

partikularna:

$$D_x(x,y) e^{-yx} = y$$
$$\rightsquigarrow D(x,y) = \int y e^{+yx}\,dx = e^{+yx} + C(y)$$

$$u = 1 + C(y) e^{-yx}$$
$$u(x,x) = 1 + C(x) e^{-x^2} = e^{-x^2} \Rightarrow C(x) = \frac{e^{-x^2} - 1}{e^{-x^2}} = 1 - e^{x^2}$$

$$u(x,y) = 1 + (1 - e^{-y^2}) e^{-yx}$$

$$u_{xy} + 2u_y = e^{-2x} \qquad \begin{cases} u(x,0) = e^{-x} \\ u_x(x,0) = -e^{-x} \end{cases} \qquad u(0,y) = 1$$

---

integriamo per y

$$\rightsquigarrow \quad u_x + 2u = ye^{-2x} + C(x)$$

$$v(x,0): \quad -e^{-x} + 2e^{-x} = C(x) = e^{-x}$$

$$u_x + 2u = ye^{-2x} + e^{-x}$$

hom:

$$\frac{\partial u}{u} = -2dx \Rightarrow u = e^{-2x} C(y)$$

part: $\quad C_x(x,y)e^{-2x} = ye^{-2x} + e^{-x}$

$$C(x,y) = \int (y + e^{x})dx = yx + e^{x} + D(y)$$

$$u = yxe^{-2x} + e^{-x} + e^{-2x} D(y)$$

$$v(0,y): \quad 0 + 1 + D(y) = 1 \Rightarrow D(y) = 0$$

$$\Rightarrow u = yxe^{-2x} + e^{-x}$$